

COURSE CODE:CPT 444

COURSE TITLE: SEED PRODUCTION AND PATHOLOGY

SEED

A seed is a living biological entity comprising of a seed coat (which is the protective cover), an endosperm or in some cases cotyledon (which serves as food reserve) and an embryo (which is the seat of life), and which when exposed to favourable conditions (i.e. moisture, oxygen and temperature) would develop into a seedling. From this definition, it can be seen that the embryo is embedded in a supporting food storage organ all of which enjoy the protection of a protective cover. The embryo is also the organ in a seed component that initiates development and produces a seedling under favourable conditions. The embryo is therefore the seed. Seed technology therefore seeks to protect this biological entity called embryo and look after its welfare, while the focus of food technology is on the second component i.e. the supporting tissue and in some cases the protective cover. A seed is therefore produced to be used as planting material while grains produced to be used for food, feed or for industrial uses.

Differences between Scientifically Produced Seed and Grain (used as seed)

1. Seed is the result of a well-planned seed programme for planting, whereas grain is planted for commercial purposes (eating, animal feed and industrial uses).
2. Seed is the result of sound scientific knowledge, organised effort, investment on processing, storage and marketing facilities, whereas no such effort or knowledge is required in grain production.
3. The pedigree of the seed is ensured as being related to the initial breeder seed, whereas the varietal purity of grains is not known.
4. Efforts are usually made during seed production to rogue out off-types, diseased plants, objectionable weeds and other crop plants at appropriate stages of crop development to ensure satisfactory seed purity and health. However, no such effort is made in grain production; hence the purity and health may be inferior.

5. Seed is scientifically processed, treated, packed and labeled with proper lot identity, the grain used as seed may be manually cleaned, in some cases prior to sowing. It may also be treated but not usually labeled.
6. Seed is tested for planting quality, i.e. germination, purity, admixture or weed and other crop seeds, seed health and seed moisture content. These are not done with grains.
7. Whereas the seed quality is usually supervised by an agency not related with production (seed certification agency), there is no quality control with grains.
8. Quality standards have already been set which seeds have to meet before they are accepted as seeds, the quality of seeds is therefore well known. No such standards apply with grains. The quality is not known.
9. The seed containers carry certification tags which serve as quality marks, while grains do not have labels certification marks.

SEED PROGRAMME

A seed programme is essentially a scheme of activities planned and implemented to secure timely supply of improved seeds of the desired quality and required quantities. The effectiveness of seed programme are however, better appreciated when other factors of production like inputs (fertilizers, herbicides, pesticides) are adequately supplied.

Types of Seed Programmes

The major objective of developing a seed programme is to improve agriculture through the supply of improved good quality seed. There are three types of seed programmes.

Official Seed Programmes

In an official seed programme, the government bears the complete responsibility of developing the programme and making good quality seed available to farmers.

The disadvantages of official seed programme are:

- (a) There is seldom any concern for return on investment, or even for recovery of capital costs.

(b) Such programmes are often subject to pressures and usually too frequent personnel changes which mar its progress.

(c) Such programmes are not very efficient.

In view of the above disadvantages, there is need for private seed programme in order to ensure the survival of seed industry.

Semi-official Seed Programme

A semi-official seed programme involves the establishment of a national agency to produce, process and distribute seed. This is a more remote form of Government participation. Such an agency may be established at the initiation of a programme or at a subsequent phase of less direct government participation. Such agencies operate as autonomous units and are usually more commercial in nature. Better managed than government own and more efficient in operation.

Private Seed Programme

Private seed enterprises are agencies that handle production and distribution of seed as a business. These are enterprises that are well managed and capable to stand competition. This kind of seed programme is common in most developed economies like the United States and Western Europe. No Government involvement. In developing countries, the best approach is to involve the three types of seed programmes (the government, semigovernment and private agencies) to ensure adequate seed delivery system.

Steps Involved in Development of a Seed Programme

Once a decision has been taken to commence a seed programme, the next step is to plan its organisation and operation chart, (Organogram chart). This is a highly technical assignment and is also time consuming. It involves the collection and analysis of pertinent data on the various requirements. It is strongly advised that targets should be precise and carefully avoid over production while preparing the blueprint. Seeds by its nature are perishable and excess production may lead to wastages. The development of a viable seed programme should therefore conform to the following steps:

Collection of Pertinent Data

The first step in the development of a seed programme is to collect data:

1. On the population of farmers, the programme is expected to serve and the number of hectorage to be planted.
2. Availability of superior varieties: It is important to know the list of available superior varieties, their areas of adaptation and expected yield. Information on farmer's preferences should also be noted. Such information is valuable so that seed programme will develop the particular variety or varieties of seed in demand.
3. Availability of other inputs: Availability of irrigation with other inputs, (fertilizer, insecticides) will encourage farmers to plant more hectares and eventually there will be higher demands in seeds.
4. Seed rate for individual crop which will help in knowing the total number of kilogrammes needed per year.
5. Seed producer should be well grounded to know when new varieties should be introduced. (Renewal period).
6. Role played by extensions agents in getting farmers aware of the existence of high yielding varieties.
7. Collection of climatological data: detailed climatological data is necessary for the preparation of a calendar of operations for seed production and processing (seed drying, processing, storage and distribution of seeds).
8. Package of agronomical practices is needed to guide farmers to produce good crops.
9. Financial resources of seed growers.

Basic Strategy of Seed Production

The following strategies should be adopted to ease seed production and marketing of high quality produce.

1. Integrated development: The production of quality seeds is facilitated when all necessary inputs along with social facilities for the well-being of the producers are made available in the production area.

2. Accessibility: The farm should be accessible to aid in transportation of goods

Advantages of Accessibility

(a) It makes it easier and more convenient to provide technical guidance, supervision and training to seed producers.

(b) The integrated development discussed earlier can be executed easily.

(c) It facilitates Seed Certification Officers in discharging their duties.

(d) Substantially lowers the overhead expenses.

3. Organisation of production: Seed production is usually organised in one of the following ways:

(a) Through selected contract growers

(b) By direct effort

1. In a contract system of seed production, profit to the seed producers is low.

2. Quality of seed produced is poor.

3. Also maintenance of the seed farms is poor In a direct effort system:

1. The production of seed is undertaken by the company herself

2. The company pays for the labor of production

3. Produce good quality seed

4. Maintenance of seed farm is good

Regardless of the system used, the selected contract growers or production personnel in the direct effort system should have technical ability and personal reliability.

4. Criteria for site selection: The following criteria must be observed in selection of site for seed production.

- (a) Cropping history of the site to be used should be known so as to avoid contamination by volunteer plants and off-types from previous crops of the same family.
 - (b) The area selected should be suitable for production of high quality seed.
 - (c) For dry season farming, areas with assured irrigation facilities should be preferred.
5. Technical guidance: Active involvement of the agricultural universities/research institutes with seed programmes should be preferred because they are important:
- (a) In developing new varieties and also in screening varieties before release
 - (b) In seed multiplication of breeders and foundation seed
 - (c) In providing technical guidance
 - (d) They can be involved in seed certification and seed testing.
6. Choice of varieties: Only superior varieties of crops adapted to specific agricultural areas should be plant

POLLINATION

In angiosperms, pollination is defined as the placement or transfer of pollen from the anther to the stigma of the same flower or another flower. In gymnosperms, pollination involves pollen transfer from the male cone to the female cone. Upon transfer, the pollen germinates to form the pollen tube and the sperm for fertilizing the egg. Pollination has been well studied since the time of Gregor Mendel. Mendel successfully carried out self- as well as crosspollination in garden peas while studying how characteristics were passed on from one generation to the next. Today's crops are a result of plant breeding, which employs artificial selection to produce the present-day cultivars. A case in point is today's corn, which is a result of years of breeding that started with its ancestor, teosinte. The teosinte that the ancient Mayans originally began cultivating had tiny seeds—vastly different from today's relatively giant ears of corn. Interestingly, though these two plants appear to be entirely different, the genetic difference between them is miniscule.

Pollination takes two forms: self-pollination and cross-pollination. Self-pollination occurs when the pollen from the anther is deposited on the stigma of the same flower, or another flower on the

same plant. Cross-pollination is the transfer of pollen from the anther of one flower to the stigma of another flower on a different individual of the same species. Selfpollination occurs in flowers where the stamen and carpel mature at the same time, and are positioned so that the pollen can land on the flower's stigma. This method of pollination does not require an investment from the plant to provide nectar and pollen as food for pollinators.

Living species are designed to ensure survival of their progeny; those that fail become extinct. Genetic diversity is therefore required so that in changing environmental or stress conditions, some of the progeny can survive. Self-pollination leads to the production of plants with less genetic diversity, since genetic material from the same plant is used to form gametes, and eventually, the zygote. In contrast, cross-pollination—or out-crossing—leads to greater genetic diversity because the microgametophyte and megagametophyte are derived from different plants.

Because cross-pollination allows for more genetic diversity, plants have developed many ways to avoid self-pollination. In some species, the pollen and the ovary mature at different times. These flowers make self-pollination nearly impossible. By the time pollen matures and has been shed, the stigma of this flower is mature and can only be pollinated by pollen from another flower. Some flowers have developed physical features that prevent selfpollination. The primrose is one such flower. Primroses have evolved two flower types with differences in anther and stigma length: the pin-eyed flower has anthers positioned at the pollen tube's halfway point, and the thrum-eyed flower's stigma is likewise located at the halfway point. Insects easily cross-pollinate while seeking the nectar at the bottom of the pollen tube. This phenomenon is also known as heterostyly. Many plants, such as cucumber, have male and female flowers located on different parts of the plant, thus making selfpollination difficult. In yet other species, the male and female flowers are borne on different plants (dioecious). All of these are barriers to self-pollination; therefore, the plants depend on pollinators to transfer pollen. The majority of pollinators are biotic agents such as insects (like bees, flies, and butterflies), bats, birds, and other animals. Other plant species are pollinated by abiotic agents, such as wind and water.

Pollination by Insects

Bees are perhaps the most important pollinator of many garden plants and most commercial fruit trees. The most common species of bees are bumblebees and honeybees. Since bees cannot see the color red, bee-pollinated flowers usually have shades of blue, yellow, or other colors. Bees

collect energy-rich pollen or nectar for their survival and energy needs. They visit flowers that are open during the day, are brightly colored, have a strong aroma or scent, and have a tubular shape, typically with the presence of a nectar guide. A nectar guide includes regions on the flower petals that are visible only to bees, and not to humans; it helps to guide bees to the center of the flower, thus making the pollination process more efficient. The pollen sticks to the bees' fuzzy hair, and when the bee visits another flower, some of the pollen is transferred to the second flower. Recently, there have been many reports about the declining population of honeybees. Many flowers will remain unpollinated and not bear seed if honeybees disappear. The impact on commercial fruit growers could be devastating.

Pollination by Bats

In the tropics and deserts, bats are often the pollinators of nocturnal flowers such as agave, guava, and morning glory. The flowers are usually large and white or pale-colored; thus, they can be distinguished from the dark surroundings at night. The flowers have a strong, fruity, or musky fragrance and produce large amounts of nectar. They are naturally large and wide-mouthed to accommodate the head of the bat. As the bats seek the nectar, their faces and heads become covered with pollen, which is then transferred to the next flower.

Pollination by Birds

Many species of small birds, such as the hummingbird and sun birds, are pollinators for plants such as orchids and other wildflowers. Flowers visited by birds are usually sturdy and are oriented in such a way as to allow the birds to stay near the flower without getting their wings entangled in the nearby flowers. The flower typically has a curved, tubular shape, which allows access for the bird's beak. Brightly colored, odorless flowers that are open during the day are pollinated by birds. As a bird seeks energy-rich nectar, pollen is deposited on the bird's head and neck and is then transferred to the next flower it visits. Botanists have been known to determine the range of extinct plants by collecting and identifying pollen from 200-year-old bird specimens from the same site.

Pollination by Wind

Most species of conifers, and many angiosperms, such as grasses, maples and oaks, are pollinated by wind. Pine cones are brown and unscented, while the flowers of windpollinated

angiosperm species are usually green, small, may have small or no petals, and produce large amounts of pollen. Unlike the typical insect-pollinated flowers, flowers adapted to pollination by wind do not produce nectar or scent. In wind-pollinated species, the microsporangia hang out of the flower, and, as the wind blows, the lightweight pollen is carried with it.

Pollination by Water

Some weeds, such as Australian sea grass and pond weeds, are pollinated by water. The pollen floats on water, and when it comes into contact with the flower, it is deposited inside the flower.

Development of the Seed

The mature ovule develops into the seed. A typical seed contains a seed coat, cotyledons, endosperm, and a single embryo

The storage of food reserves in angiosperm seeds differs between monocots and dicots. In monocots, such as corn and wheat, the single cotyledon is called a scutellum; the scutellum is connected directly to the embryo via vascular tissue (xylem and phloem). Food reserves are stored in the large endosperm. Upon germination, enzymes are secreted by the aleurone, a single layer of cells just inside the seed coat that surrounds the endosperm and embryo. The enzymes degrade the stored carbohydrates, proteins and lipids, the products of which are absorbed by the scutellum and transported via a vasculature strand to the developing embryo. Therefore, the scutellum can be seen to be an absorptive organ, not a storage organ.

The two cotyledons in the dicot seed also have vascular connections to the embryo. In endospermic dicots, the food reserves are stored in the endosperm. During germination, the two cotyledons therefore act as absorptive organs to take up the enzymatically released food reserves, much like in monocots (monocots, by definition, also have endospermic seeds). Tobacco (*Nicotiana tabacum*), tomato (*Solanum lycopersicum*), and pepper (*Capsicum annuum*) are examples of endospermic dicots. In non-endospermic dicots, the triploid endosperm develops normally following double fertilization, but the endosperm food reserves are quickly remobilized and moved into the developing cotyledon for storage. The two halves of a peanut seed (*Arachis hypogaea*) and the split peas (*Pisum sativum*) of split pea soup are individual cotyledons loaded with food reserves.

The seed, along with the ovule, is protected by a seed coat that is formed from the integuments of the ovule sac. In dicots, the seed coat is further divided into an outer coat known as the testa and inner coat known as the tegmen.

The embryonic axis consists of three parts: the plumule, the radicle, and the hypocotyl. The portion of the embryo between the cotyledon attachment point and the radicle is known as the hypocotyl (hypocotyl means “below the cotyledons”). The embryonic axis terminates in a radicle (the embryonic root), which is the region from which the root will develop. In dicots, the hypocotyls extend above ground, giving rise to the stem of the plant. In monocots, the hypocotyl does not show above ground because monocots do not exhibit stem elongation.

The part of the embryonic axis that projects above the cotyledons is known as the epicotyl. The plumule is composed of the epicotyl, young leaves, and the shoot apical meristem.

Upon germination in dicot seeds, the epicotyl is shaped like a hook with the plumule pointing downwards. This shape is called the plumule hook, and it persists as long as germination proceeds in the dark. Therefore, as the epicotyl pushes through the tough and abrasive soil, the plumule is protected from damage. Upon exposure to light, the hypocotyl hook straightens out, the young foliage leaves face the sun and expand, and the epicotyl continues to elongate. During this time, the radicle is also growing and producing the primary root. As it grows downward to form the tap root, lateral roots branch off to all sides, producing the typical dicot tap root system.

In monocot seeds, the testa and tegmen of the seed coat are fused. As the seed germinates, the primary root emerges, protected by the root-tip covering: the coleorhiza. Next, the primary shoot emerges, protected by the coleoptile: the covering of the shoot tip. Upon exposure to light (i.e. when the plumule has exited the soil and the protective coleoptile is no longer needed), elongation of the coleoptile ceases and the leaves expand and unfold. At the other end of the embryonic axis, the primary root soon dies, while other, adventitious roots (roots that do not arise from the usual place — i.e. the root) emerge from the base of the stem. This gives the monocot a fibrous root system.

Seed Germination

Many mature seeds enter a period of inactivity, or extremely low metabolic activity: a process known as dormancy, which may last for months, years or even centuries. Dormancy helps keep

seeds viable during unfavorable conditions. Upon a return to favorable conditions, seed germination takes place. Favorable conditions could be as diverse as moisture, light, cold, fire, or chemical treatments. After heavy rains, many new seedlings emerge. Forest fires also lead to the emergence of new seedlings. Some seeds require vernalization (cold treatment) before they can germinate. This guarantees that seeds produced by plants in temperate climates will not germinate until the spring. Plants growing in hot climates may have seeds that need a heat treatment in order to germinate, to avoid germination in the hot, dry summers. In many seeds, the presence of a thick seed coat retards the ability to germinate. Scarification, which includes mechanical or chemical processes to soften the seed coat, is often employed before germination. Presoaking in hot water, or passing through an acid environment, such as an animal's digestive tract, may also be employed.

Depending on seed size, the time taken for a seedling to emerge may vary. Species with large seeds have enough food reserves to germinate deep below ground, and still extend their epicotyl all the way to the soil surface. Seeds of small-seeded species usually require light as a germination cue. This ensures the seeds only germinate at or near the soil surface (where the light is greatest). If they were to germinate too far underneath the surface, the developing seedling would not have enough food reserves to reach the sunlight.

Development of Fruit and Fruit Types

After fertilization, the ovary of the flower usually develops into the fruit. Fruits are usually associated with having a sweet taste; however, not all fruits are sweet. Botanically, the term "fruit" is used for a ripened ovary. In most cases, flowers in which fertilization has taken place will develop into fruits, and flowers in which fertilization has not taken place will not. Some fruits develop from the ovary and are known as true fruits, whereas others develop from other parts of the female gametophyte and are known as accessory fruits. The fruit encloses the seeds and the developing embryo, thereby providing it with protection. Fruits are of many types, depending on their origin and texture. The sweet tissue of the blackberry, the red flesh of the tomato, the shell of the peanut, and the hull of corn (the tough, thin part that gets stuck in your teeth when you eat popcorn) are all fruits. As the fruit matures, the seeds also mature.

Fruits may be classified as simple, aggregate, multiple, or accessory, depending on their origin. If the fruit develops from a single carpel or fused carpels of a single ovary, it is known as a simple fruit, as seen in nuts and beans. An aggregate fruit is one that develops from more than one carpel, but all are in the same flower: the mature carpels fuse together to form the entire fruit, as seen in the raspberry. Multiple fruit develops from an inflorescence or a cluster of flowers. An example is the pineapple, where the flowers fuse together to form the fruit. Accessory fruits (sometimes called false fruits) are not derived from the ovary, but from another part of the flower, such as the receptacle (strawberry) or the hypanthium (apples and pears).

There are four main types of fruits. Simple fruits, such as these nuts, are derived from a single ovary. Aggregate fruits, like raspberries, form from many carpels that fuse together. Multiple fruits, such as pineapple, form from a cluster of flowers called an inflorescence. Accessory fruit, like the apple, are formed from a part of the plant other than the ovary.

Fruits generally have three parts: the exocarp (the outermost skin or covering), the mesocarp (middle part of the fruit), and the endocarp (the inner part of the fruit). Together, all three are known as the pericarp. The mesocarp is usually the fleshy, edible part of the fruit; however, in some fruits, such as the almond, the endocarp is the edible part. In many fruits, two or all three of the layers are fused, and are indistinguishable at maturity. Fruits can be dry or fleshy.

Furthermore, fruits can be divided into dehiscent or indehiscent types. Dehiscent fruits, such as peas, readily release their seeds, while indehiscent fruits, like peaches, rely on decay to release their seeds.

Fruit and Seed Dispersal

The fruit has a single purpose: seed dispersal. Seeds contained within fruits need to be dispersed far from the mother plant, so they may find favorable and less competitive conditions in which to germinate and grow.

Some fruit have built-in mechanisms so they can disperse by themselves, whereas others require the help of agents like wind, water, and animals.

Modifications in seed structure, composition, and size help in dispersal. Wind-dispersed fruit are lightweight and may have wing-like appendages that allow them to be carried by the wind. Some

have a parachute-like structure to keep them afloat. Some fruits—for example, the dandelion—have hairy, weightless structures that are suited to dispersal by wind.

Seeds dispersed by water are contained in light and buoyant fruit, giving them the ability to float. Coconuts are well known for their ability to float on water to reach land where they can germinate. Similarly, willow and silver birches produce lightweight fruit that can float on water.

Animals and birds eat fruits, and the seeds that are not digested are excreted in their droppings some distance away. Some animals, like squirrels, bury seed-containing fruits for later use; if the squirrel does not find its stash of fruit, and if conditions are favorable, the seeds germinate. Some fruits, like the cocklebur, have hooks or sticky structures that stick to an animal's coat and are then transported to another place. Humans also play a big role in dispersing seeds when they carry fruits to new places and throw away the inedible part that contains the seeds.

All of the above mechanisms allow for seeds to be dispersed through space, much like an animal's offspring can move to a new location. Seed dormancy, which was described earlier, allows plants to disperse their progeny through time: something animals cannot do. Dormant seeds can wait months, years, or even decades for the proper conditions for germination and propagation of the species.

Fruits and seeds are dispersed by various means. (a) Dandelion seeds are dispersed by wind, the (b) coconut seed is dispersed by water, and the (c) acorn is dispersed by animals that cache and then forget it.

SEED STORAGE

After harvest, seed has a transition period, which is from time of harvest to the time of planting or consumption. This transition period is referred to as time of storage. Storage could be long or short depending upon the circumstances and the use to which the seed is intended. Whatever the intended usage may be, seed storage is a very important period that can lead to either safety or deterioration of seed. Care must be taken to store seeds according to prescribed guidelines. General principles of seed storage The entire storage life of seed has been conveniently divided into the following stages:

1. Storage of seed plant in the field: Once seed has attained physiological maturity and about to be harvested on the field, any delay in harvest means that the seed has been stored on the field. The

seed is therefore exposed to adverse weather conditions and attack by fungi, insects, birds and rodents.

2.From period of harvest until processing: This transition period between harvest and processing may take long or short period depending on individual crop. This storage period is referred to as 'after-ripening'. While it is desirable for some crops to attain full maturity, it may be disastrous for some crops if it goes beyond certain limits. Some farmers leave the seed stored in the open on the farm while others store seed in a warehouse pending threshing.

3.Storage in warehouses: This is the storage period that follows after seed has been processed and packaged. This is considered to be the longest period of storage because seed stay in this state until they are distributed for either planting or further processing into food, feed or any other industrial use in case of grain.

4.Storage while in transit: After storage in warehouse, seed is moved either to the farm directly for distribution to farmers, or to the industry in case of grain. Seed is therefore stored during this transit period in rail wagons, trucks, carts, rail sheds and boots of cars.

5.Storage in retail shops: Seed meant to be distributed to farmers sometimes get into market retail shops.

6.Storage on the user's farm: sometimes, not the total stock of seed conveyed to the farm can be planted same day. The rest seed can be stored in a farm storehouse pending when it will be put to use.

Factors Affecting Seed Longevity in Storage

Seed longevity refers to the period during which a seed remains viable and can germinate when exposed to favourable conditions until when viability is lost. This period varies widely in different crops. Some crops are short-lived and therefore have short storage life while some may remain viable for decades. Seed ageing and loss of germination during storage cannot be stopped altogether but it can be appreciably reduced by providing good storage conditions. Factors that affect the life span of seed during storage include:

1.Kind/variety of the seed: The seed of some varieties are naturally short-lived while other may have longer shelf life in storage due to differences in their genetic constitution.

2.Initial seed quality: Clean undeteriorated seed with high vigour store longer than seed which had started deteriorating due to weathering damage, mechanical injuries, wrinkled or other causes. It is therefore advisable that only high quality seed should be stored.

3.Moisture content: High seed moisture content greatly accelerates seed deterioration and its life span. Recommended moisture contents for safe seeds should be adhered to.

4.Relative humidity and temperature: Temperature and humidity are the two most important factors affecting seed quality in storage. The higher the temperature of the seed, the lower the life span. Also the higher the humidity of the seed the lower the life span.

5.Rodents and birds: All openings on the storage structure should be sealed or screened against birds, rats and other rodents. Adequate ventilation should be provided since they are constant source of seed loss.

6.Storage in transit, at the retailer's store and user's farm: All effort about construction of a befitting warehouse would be lost if in the end seed would lose its viability either while in transit, retailer's store or on the user's farm house. Adequate storage precautions at these points are therefore very necessary.

The general principles that are necessary for good storage are:

1. Seed storage should be in dry and cool conditions
- 2.Seed storage should have effective pest control facilities.
- 3.There should be proper sanitation in seed stores.
4. Seed should be dried to safe moisture limits before being kept in the store.
- 5.Only well cleaned, treated seed of high germination and vigour should be stored.
- 6.It is necessary to determine seed storage needs in terms of length of storage, and prevailing climate of the area during storage period. Regions with favorable storing conditions (less humidity and dry air) require less sophisticated building materials compared to areas with high humidity.

SEED PATHOLOGY

Seeds are essential for maintenance of life. Seeds not only sustain man and animals, they are also the source of food for multitudes of micro-organisms living in the ecosystem.

Seed pathology is the study of diseases that affect seeds, significantly influencing crop yield, quality, and ultimately food security. Seed-borne diseases are responsible for substantial agricultural losses, making the study of seed pathology crucial for sustainable agriculture.

Terminologies;

Pathogen: An entity, usually a micro-organism that can incite disease. In a literal sense a pathogen is any agent that causes pathos (ailment, suffering) or damage. However, the term is generally used to denote living organisms (Fungi, bacteria, nematodes etc.) and viruses but not nutritional deficiencies.

Pathogenicity is the ability of the pathogen to cause disease.

Pathogenesis is the chain of events that lead to development of disease in the host (or) sequence of progress in disease development from the initial contact between the pathogen and its host to the completion of the syndrome.

Types of Seed-Borne Pathogens Fungi: Examples: *Aspergillus*, *Fusarium*, *Alternaria*.

Characteristics: These pathogens typically reproduce through spores and can infect seeds both externally (on the seed surface) and internally (within the embryo or endosperm). Fungal pathogens are often responsible for seed decay, mold formation, and reduced germination.

Bacteria: Examples: *Xanthomonas*, *Pseudomonas*, *Erwinia* **Characteristics:** Bacterial pathogens are single-celled organisms with structures like cell walls and flagella. They can also infect seeds externally or internally, causing symptoms such as seed rot and discoloration.

Viruses: Examples: Tobacco mosaic virus (TMV), Tomato spotted wilt virus (TSWV).

Characteristics: Viruses consist of a protein coat surrounding a nucleic acid (RNA or DNA). These pathogens can be transmitted through seeds either directly or via vectors like insects, leading to reduced seedling vigor and growth abnormalities.

Nematodes: Examples: *Meloidogyne*, *Heterodera* **Characteristics:** These are microscopic, wormlike organisms that lay eggs and can cause significant damage to seeds by penetrating and

feeding on plant tissues, often leading to poor seedling establishment. Seed Infection and Disease Cycle

Infection Sources: Seed-borne diseases can originate from contaminated soil, water, air, insects, or equipment.

Infection Modes: External: Colonization occurs on the seed surface. Internal: Infection may occur within the seed's internal structures, including the embryo or endosperm.

Disease Cycle: Typically involves four stages: Infection: Pathogens enter the seed. Colonization: Pathogens spread within the seed. Reproduction: Pathogens reproduce within the seed. Dissemination: Pathogens spread to other seeds or plants.

Symptoms of Seed-Borne Diseases: Visible Symptoms: Discoloration, Mold growth, Seed rot Physical deformation of seeds Invisible Symptoms: Reduced germination rate, Lowered seed vigor, Poor seedling growth.

Diagnosis of Seed-Borne Diseases: Visual Examination: The first step in diagnosing seed-borne diseases. Laboratory Tests: Culturing: Growing the pathogen on specific media. Serology: Detecting pathogens using antibodies. PCR (Polymerase Chain Reaction): Amplifying pathogen DNA for identification. ELISA (EnzymeLinked Immunosorbent Assay): Detecting specific pathogen proteins. Seed Health Testing: Blotter Test: Incubating seeds on blotter paper to detect fungal growth. Agar Test: Culturing seeds on agar media. Seedling Grow-Out Test: Observing seedling growth under controlled conditions. Management of Seed-Borne Diseases

Seed Selection and Certification: Using certified disease-free seeds.

Seed Treatment: Chemical: Fungicides and bactericides. Physical: Heat treatment or irradiation. Biological: Using antagonistic microorganisms to control pathogens.

Crop Rotation and Sanitation: Preventing disease buildup by rotating crops and maintaining clean equipment.

Integrated Pest Management (IPM): Combining biological, chemical, and cultural practices to manage seed-borne diseases effectively.

Since most of the cultivated crops are raised from seed the transmission of diseases and transport of pathogens has much importance. The dormant structures of the pathogen (Ex: seeds of *Cuscuta*, *Sclerotia* of ergot fungus, smut sori, etc.) are found mixed with seed lots and they are dispersed as seed contaminants. The bacterial cells or spores of fungi present on the seed coat (such as in smuts of barley, sorghum, etc.) are transported to long distances. Dormant mycelium of many fungi present in the seed is transmitted to long distances. There are three types of dispersal by seed, viz., contamination of the seed, externally seed borne and internally seed borne.

a) Contamination of the seed: Seed borne pathogens move in seed lot as separate contaminants without being in intimate contact with the viable crop seeds. The seeds of the pathogen or parasite and the host are mixed during harvest of the crop. In many cases, the identity of the seeds of the two entities (host and the pathogens) is difficult to separate. Ex: Smut of pearl millet and ergot of rye. Smut sori and ergots mix easily with the seed lots during harvest and threshing.

b) Externally seed borne: Close contact between structure of the pathogen and seeds is established where the pathogen gets lodged in the form of dormant spores or bacteria on the seed coat during growth of the crop or at the time of harvest and threshing. Ex: Short smut of sorghum, bacterial blight of cotton, loose smut of barley etc.

In many pathogens the externally seed borne structures such as smut spores can persist for many years due to their inherent capacity for long survival.

c) Internally seed borne: The pathogen may penetrate into the ovary and cause infection of the embryo while it is developing. They become internally seed borne. Ex: Loose smut of wheat.